

Anirudh Penmatcha

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Education

Master of Science - Data Science

August, 2023 - May, 2025

Indiana University, Bloomington

Indiana, USA

Coursework : Cloud Comp., Adv. DB Concepts, App. DB Tech, Algorithms, Data Mining, ML for Econ Data, Graph Analytics, Statistics

Bachelor of Technology - Computer Science

August, 2018 - May, 2022

SRM Institute of Science and Technology, Chennai

Chennai, India

Coursework : Computer Networking, Operating System, Database Management Systems, Adv. Programming Practice, Distributed Systems

Work Experience

Enterprise Data Analyst - Coop

September, 2024 - December, 2024

Yazaki North America

Remote

- Implemented a solution to address the **business problem** of finding replacements for wire harness' components.
- Developed **MSSQL** queries on data with over **26 million** rows utilizing CTEs, CTAS, Functions, and Procedures.
- Applied several query optimization techniques, achieving a **90% reduction in runtime**.
- Created **Power BI** visualizations to review and validate results prior to **staging implementation**.
- Further optimized code for **distributed data storage and computing**, ensuring seamless deployment into production.
- Comprehensively documented high-level and low-level logic, **successfully delivering all project objectives**.

Applied Machine Learning Intern

June, 2024 - August, 2024

Yazaki North America

Michigan, USA

- Developed complex **SQL** queries to extract & manage data subsets with over **300,000** records from a Data Lakehouse.
- Performed data cleaning, transformation, and visualization to **prepare complex datasets** for analysis.
- Applied advanced **Clustering** and **NLP** techniques, including **K-Means**, **DBScan**, **Cosine**, and **Gower's similarity**.
- Visualized insights in **Power BI**, enhancing data-driven decisions, reducing manufacturing time and costs.

Data Science Intern

September, 2021 - February, 2022

Practically

Telangana, India

- Developed a pipeline to extract questions using **TensorFlow & OpenCV** from a dataset of over 30,000 real test papers.
- Achieved 15% improvement in system performance by **optimizing** the code and pipeline.
- Maintained **detailed documentation** and communicated progress regularly to key stakeholders.

Projects

A Study on the Scalability of Big Data Analytics with Dask

- Created a Dask based cluster in **AWS** using **Docker** that can auto-scale and maintain persistent storage.
- Manually setup the network configuration and shared data drive using **Elastic File System (EFS)**.
- Built and deployed docker images within the **Elastic Container Repository (ECR)**.

Autonomous System Researcher - RUDRA

- Built object detection, autonomous navigation and intersystem communications using **Python**, **TensorFlow & ROS**.
- Created a **Gazebo** simulation to evaluate traversal algorithms, alongside the development of **AR tag detection**.
- Led a team in overhauling the **autonomous system stack**, and recruited, trained, and supervised team members.

Technical Skills

Languages : Python, SQL (Microsoft SQL Server, PostgreSQL, MySQL)

Tools : Visual Studio, PyCharm, Azure Data Studio

ML Algorithms : Regression, Classification, Clustering, NN, CV, NLP